Geometric motivic integration : introduction and motivation

Developed to study the relation between additive invariants of varieties (→ Betti or Hodge numbers) in the context of Mirror Symmetry in String Theory

↳ Main objects of study are Calabi-Yau varieties, i.e. compact, complex algebraic varieties admitting a non-vanishing form of maximal degree

(1) Batyrev (1995): birationally equivalent Calabi-Yau varieties have the same Betti-numbers.

↳ Hironaka
  ↳ Common smooth birational model
↳ Reduction mod p^m and Weil's conjectures:
  zeta function associated with counting points mod p^m on smooth varieties
  ↳ Strong results by Dwork, Grothendieck and Deligne
↳ p-adic integration:
  "change of variables formula"
  ↳ Relate different p-adic volumes
↳ Comparison theorem between ℓ-adic cohomology ($\ell \neq p$) and usual Betti-numbers

(2) Kontsevich (1995, Orsay): more direct approach avoiding p-adic integration and Weil's conjectures

↳ arc spaces ($\mathbb{C}[[t]]$ "t-adic" spaces) as domain of integration
↳ measure based on a universal additive invariant of varieties (Grothendieck ring of complex varieties $K_0(Var_{\mathbb{C}})$)
↳ change of variables formula
  ↳ "jacobian" expressed in terms of contact order of arcs along divisors
  ↳ also same Hodge numbers (under same hypothesis as Batyrev)

⇒ Motivic integration

(3) Further development by Denef and Loeser, and Looijenga
  ↳ Batyrev: "stringy" invariants (applications to singularities)
  ↳ Cluckers and Loeser: model theory (general framework)

Zeta functions and monodromy conjecture

Application to singularities

(1) I. Gel'fand (1954, I(M)): meromorphic continuation of

$$Z(f,\varphi;s) = \int_{\mathbb{R}^n} |f(x)|^s \varphi(x)\, dx$$

↳ "test function": C^∞-function with compact support

(2) Two solutions ('70):

- Bernstein and S. Gel'fand, and Atiyah using resolutions of singularities
- Bernstein using b-function
  ↳ Bernstein-Sato polynomial

(3) <u>Theorem</u>: (Kashiwara, Malgrange, 1983): poles of $Z(f,\varphi,s)$ corresponds to monodromy eigenvalues of $f^{-1}(0)$.
Topology of the Milnor fibre

(4) p-adic Igusa zeta function (introduced by Weil, studied by Igusa):

$$Z_p(f,\varphi;s) = \int_{\mathbb{Q}_p^n} |f(x)|_p^s \varphi(x)\, dx$$

↗ locally constant with compact support

↳ rational function on p^{-s}

↳ Monodromy conjecture (1988)

↳ Denef's formula using resolution of singularities

↳ The topological zeta function $Z_{top}(f;s)$ (D-L, 1991): "limit" of $Z_p(f,\varphi;s)$

↳ The motivic zeta function $Z_{mot}(f;s)$ (D-L, 1998 and 1999): $Z_p(f,\varphi;s)$ and $Z_{top}(f;s)$ are specializations of $Z_{mot}(f;s)$

More geometric way to understand zeta functions and connection to the monodromy eigenvalues